EXP.NO:25

Pair with Difference-O(n)Time Complexity, O(1) Space Complexity

PROBLEM STATEMENT:

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

EXP.NO:25

CODE

```c
#include <stdio.h>

#define MAX 100001

int main() {
    int n;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    int key;
    scanf("%d", &key);

    int seen[MAX] = {0};

    for (int i = 0; i < n; i++) {
        int required = a[i] - key;

        if (required >= 0 && required < MAX && seen[required]) {
            printf("1");
            return 0;
        }

        if (a[i] >= 0 && a[i] < MAX) {
            seen[a[i]] = 1;
        }
    }

    printf("0");
}

```

EXP.NO:25

OUTPUT:



	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RESULT:

The code is executed successfully.